Math 9 Fall 2022 — Lecture #6 Worksheet
Corresponding Outcomes: EO3, EO4, GOMa1, GOMa2

This is an in-class worksheet. You do not need to turn in anything.

1. Write a function `myrow(A)` that takes as input a matrix and as output returns the 2nd-to-last row of the matrix. For example, if the input if [1,2;3,4;5,6,;7,8], then the output should be [5,6].

2. Write a function `diagzero(n)` which takes as input a positive integer n and as output returns an n-by-n matrix with 0s down the diagonal and with 1s everywhere else. For example, if the input is 3, the output should be the matrix [0,1,1;1,0,1;1,1,0].

3. Write a function `leadzero(v)` which takes as input a vector v and as output returns the number of zeros at the beginning of v. for example, for input -5:5, the output would be 0. If the input is [0,0,3,0,0,0] the output would be 2. If the input is [0,0,0] the output would be 3.

4. Using a while loop, write a function nextinteger(v) which takes as input a vector v and as output returns the smallest positive integer which does not appear in v. For example, if the input is [1,2,3], the output would be 4. If the input is [4,0,1,-10], the output would be 2.

5. Using a while loop, write a function `twodigits(y)` that takes as input a real number y and as output returns the first number x in the sequence $.00, .01, .03, \ldots, 0.98, 0.99, 1, 1.01, 1.02, \ldots$, for which $x^3 - x - 1 > y$.

6. Write a function `smallrange(v)` which takes as input a row vector v, and as output returns true if there exists a number c such that all entries in v are contained in the half-open interval $[c, c+1)$, and returns false otherwise. For example, if the input is $[2.4999, 2, 2.3, 1.5]$, then the output would be true. If the input is $[7, 8]$, then the output would be false.